DoF Analysis for (M, N)-Channels through a Number-Filling Puzzle

Yue Bi, Michèle Wigger, Yue Wu, and Cunqing Hua

Abstract

We propose upper and lower bounds on the sum degrees of freedom (Sum-DoF) of K-user interference networks with general connectivity and message flows. The derivation of the upper bound is based on combining MAC-type arguments, and the resulting bound is formulated as a linear program. The lower bound is obtained by relating the Sum-DoF achieved by certain interference alignment (IA) schemes to the score of any valid solution of a number-filling puzzle that depends on the connectivity and message flow of the network. We show that finding the optimal puzzle solution (the one with the highest score) is equivalent to solving a 0-1 integer optimization problem.

For a class of networks with regular connectivity and message flows, we present explicit constructions of feasible puzzle solutions and show that their score is optimal. In some cases, it even achieves the derived upper bound. Our results on regular networks also allow us to show that a previously proposed coding scheme for a full-duplex MapReduce wireless distributed computing system with regular file assignment and computation load $\mathsf{r} = 1$ is optimal among the class of IA schemes considered in this work.

Index Terms

Degrees of freedom, wireless channel, interference alignment

I. Introduction

The sum degrees of freedom (Sum-DoF) of a network characterizes the pre-log approximation of its sum-capacity in the asymptotic regime of infinite Signal-to-Noise Ratios (SNR) [1], i.e., when the network operates in the interference-limited regime. The study of the Sum-DoF has a rich history. Classical multi-user channels include interference channels [2]–[4], where each transmitter conveys a message only to its dedicated receiver; X-channels [5]–[9], where each transmitter sends an independent message to each receiver; and cognitive or cooperative interference networks [10]–[13], where messages or signals can be shared among multiple transmitters and/or receivers. Notably, it has been demonstrated that the Sum-DoF of a fully-connected K-user IC without cooperation is $\mathsf{K}/2$ when the channel coefficients are independent and identically distributed (i.i.d.) fading according to a continuous distribution [6]. Furthermore, the Sum-DoF of the corresponding X-channel is $\mathsf{K}^2/(2\mathsf{K}-1)$. Both of these Sum-DoFs are achieved through interference alignment (IA) schemes [3], [7], [11] and apply to fully-connected interference networks where each receive signal is interfered by each transmit signal.

For general network connectivities and message flows the Sum-DoF remains unknown. In fact, even the largest Sum-DoF that can be achieved with IA schemes is unknown. In the non-cooperative setting, where each message is available at exactly one transmitter, which is the focus of this work, determining the largest achievable Sum-DoF with IA schemes amounts to finding the optimal assignment of precoding matrices to the messages. Depending on the network connectivity and message flow, the standard IA scheme with the precoding matrices proposed in [7] is generally suboptimal; see, for example, [14]. Previous studies have linked the optimal choice of IA precoding matrices for a given network topology to index coding problems [15]. More broadly, [16] established a connection between index coding and IA with combinatorial coding.

In this work, we formalize a number-filling puzzle adapted to the connectivity and the message flow of the network, and show that any valid solution for this puzzle corresponds to a valid IA scheme, with the puzzle filling determining the precoding matrices to be used in the IA scheme. The Sum-DoF achieved by this IA scheme can directly be obtained as a score from the puzzle filling. This allows future works to skip the cumbersome task of checking the alignment and separability conditions at all receivers and finding the largest Sum-DoF achieved using IA by simply finding an optimal score solution for the puzzle. We show that finding such an optimal solution is equivalent to solving a 0-1 integer program.

The considerations in the above paragraph naturally provide a lower bound on the maximum achievable Sum-DoF over a given interference network. We also propose an upper bound on the Sum-DoF by carefully combining MAC-type arguments and phrase the obtained bound in terms of a linear program minimization problem. Any feasible solution of this linear program provides a (relaxed) upper bound on the Sum-DoF.

We analyze the proposed upper and lower bounds more carefully for a class of networks with symmetric and regular network connectivity. Specifically, we propose a valid solution for the puzzle and show that it achieves the optimal puzzle score, and moreover, it is either close to or equal to the upper bound.

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A special case of the considered regular networks arises naturally during the shuffle phase of a full-duplex wireless MapReduce distributed computing system [17]–[20] under a regular file assignment to the nodes. The results in this paper show that the IA scheme proposed in [19] for the special case of computation load $r = 1$ has an optimal puzzle score, and thus achieves the largest Sum-DoF during the shuffle phase among all IA schemes considered in the framework of this paper.

To summarize, the contributions of this paper are as follows.

- For a given network connectivity and message flow, we establish a connection between IA schemes and a number-filling puzzle, where a well-defined score of the puzzle equals the Sum-DoF achieved by the corresponding IA scheme.
- We show that finding the optimal filling of the puzzle (the one with the largest score), for given connectivity and message flow, is equivalent to solving a standard 0-1 optimization problem, for which efficient numerical solvers exist.
- We derive an upper bound on the largest achievable Sum-DoF by *any* scheme. This upper bound is phrased as a linear program minimization problem, and thus any feasible solution of the linear program yields a relaxed upper bound on the Sum-DoF.
- For a class of symmetric regular networks, we provide an explicit construction of the puzzle filling and derive in closed-form the corresponding score (and thus Sum-DoF). The proposed construction: improves upon the straightforward implementation of the standard IA; coincides with the values obtained by our numerical integer programming algorithm, identifying the optimum puzzle score; and performs close to the converse bound (evaluated numerically).
- For wireless MapReduce distributed computing systems with computation load $\mathsf{r} = 1$ and regular file assignment, we show that the IA scheme proposed in [19] achieves the optimal Sum-DoF among all schemes within the IA framework considered in this paper.

The remainder of this paper is organized as follows. Section II formally introduces the communication setup and formulates the Sum-DoF problem. The following Section III introduces the corresponding number filling puzzle and presents our main results on the Sum-DoF of interference networks. Section IV proves the key proposition for the general lower bound by constructing IA schemes from valid precoding index matrices. Section V proves the converse result on Sum-DoF. Sections VI provides the achievability and converse proofs for the considered regular networks. Finally, Section VII concludes the paper.

Notations: We use sans serif font for constants, bold for vectors and matrices, and calligraphic font for most sets. The sets of complex numbers and nonnegative integers are denoted $\mathbb{C}$ and $\mathbb{N}_0$. For a finite set $\mathcal{A}$, let $|\mathcal{A}|$ denote its cardinality. For any $n \in \mathbb{Z}^+$, define $[n] \triangleq \{1, 2, \ldots, n\}$. The operation mod_1 represents modulo with an offset of 1, equivalent to regular modulo mod for a nonzero remainder, and it returns the divisor if the remainder is zero. For any vector $\mathbf{v}$, let $\mathrm{diag}(\mathbf{v})$ be the diagonal matrix with diagonal entries given by the elements of the vector $\mathbf{v}$. When writing $[\boldsymbol{v}_i \colon i \in \mathcal{S}]$ or $[\boldsymbol{v}_i]_{i\in\mathcal{S}}$ we mean the matrix consisting of the set of columns $\{\mathbf{v}_i\}_{i\in\mathcal{S}}$. For a matrix $\mathbf{A} \in \mathbb{R}^{(n\times n)}$ and finite sets $\mathcal{A}$, $\mathcal{B} \subseteq \mathbb{Z}$, we denote $\mathbf{A}[\mathcal{A}, \mathcal{B}]$ as the submatrix of $\mathbf{A}$ obtained by removing rows in $[n]\backslash(\mathcal{A} \ \mathsf{mod}_1 \ n)$ and columns in $[n]\backslash(\mathcal{B} \ \mathsf{mod}_1 \ n)$. Let further $\mathbf{A}[\mathcal{A}, :] \triangleq \mathbf{A}[\mathcal{A}, [n]]$, and $\|\mathbf{A}\|_0$ denote the number of nonzero elements in $\mathbf{A}$. For two matrices $\mathbf{A}$, $\mathbf{B} \in \{0, 1\}^{(n\times n)}$, the relationship $\mathbf{A} \subseteq \mathbf{B}$ holds if and only if $\mathbf{A}$ can be obtained by replacing certain entries in $\mathbf{B}$ with 0.

II. Channel model

Consider a network with K Txs and K Rxs labeled from 1 to K. The connectivity of the network is described by matrix $\mathbf{N} \in \{0, 1\}^{\mathsf{K}\times\mathsf{K}}$. In this channel, each Rx p observes a linear combination of the signals sent by the Txs with $\mathbf{N}[p, q] = 1$ corrupted by Gaussian noise. Denoting Tx q's slot-t input by $X_q(t) \in \mathbb{C}$ and Rx p's slot-t output by $Y_p(t) \in \mathbb{C}$, the input-output relation of the network is expressed as:

$$Y_p(t) = \sum_{\{q\colon \mathbf{N}[p,q]=1\}} H_{p,q}(t)X_q(t) + Z_p(t), \qquad p \in [\mathsf{K}], \tag{1}$$

where the sequences of complex-valued channel coefficients $\{H_{p,q}(t)\}$ and standard circularly symmetric Gaussian noises $\{Z_p(t)\}$ are both i.i.d. and independent of each other and of all other channel coefficients and noises. Each coefficient $H_{p,q}(t)$ has independent and identically distributed real and imaginary parts, following a specified continuous distribution over a bounded interval $[-\mathsf{H}_{\max}, \mathsf{H}_{\max}]$. Importantly, these coefficients are known to all terminals even before communication commences.

The message flow in this channel is represented by matrix $\mathbf{M} \in \{0, 1\}^{\mathsf{K}\times\mathsf{K}}$, in the sense that Tx q transmits an independent message $a_{p,q}$ to each Rx p for which $\mathbf{M}[p, q] = 1$. When communication is of blocklength T, each message is uniformly distributed over $\left[2^{\mathsf{T}\mathsf{R}_{p,q}}\right]$, where $\mathsf{R}_{p,q} \geq 0$ denotes the rate of transmission, and it is independent of all other messages and of all channel coefficients and noise sequences. As a consequence, Tx $q \in [\mathsf{K}]$ produces its block of channel inputs $\mathbf{X}_q \triangleq (X_q(1), \ldots, X_q(\mathsf{T}))$ as

$$\mathbf{X}_q = f_q^{(\mathsf{T})}\left(\{a_{p,q}\colon \mathbf{M}[p, q] = 1\}\right) \tag{2}$$

by means of an encoding function $f_q^{(\mathsf{T})}$ on appropriate domains, and so that the inputs satisfy the block-power constraint

$$\frac{1}{\mathsf{T}}\sum_{t=1}^{\mathsf{T}} \mathbb{E}\left[|X_q(t)|^2\right] \leq \mathsf{P}, \qquad q \in [\mathsf{K}]. \tag{3}$$

According to the definitions above, the channel is mostly described by the connectivity matrix $\mathbf{N}$ and the message flow matrix $\mathbf{M}$. For example, the two matrices

$$\mathbf{M} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{N} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \tag{4}$$

represent a standard 3-user interference channel where each Tx $q \in \{1, 2, 3\}$ sends a message only to its corresponding Rx, and each Rx observes a linear combination of all the transmit signals corrupted in additive white Gaussian noise.

Our main interest in this paper is in the sum of rates that are achievable in the high-SNR regime, either focusing only on IA schemes or also on any kind of schemes, i.e., on the high-SNR sum-capacity.

Given a power $\mathsf{P} > 0$, a rate tuple $(\mathsf{R}_{p,q} \colon p, q \in [\mathsf{K}],\ \mathbf{M}[p,q] = 1)$ is called *achievable* if for each blocklength T there exist encoding functions $\{f_q^{(\mathsf{T})}\}_{q \in [\mathsf{K}]}$ as described above and decoding functions $\{g_{p,q}^{(\mathsf{T})}\}$ on appropriate domains producing the guesses

$$\hat{a}_{p,q} = g_{p,q}^{(\mathsf{T})}(Y_p(1), \ldots, Y_p(\mathsf{T})), \quad p, q \in [\mathsf{K}],\ \mathbf{M}[p,q] = 1, \tag{5}$$

so that the sequence of error probabilities

$$p^{(\mathsf{T})}(\text{error}) \triangleq \Pr\Bigg[\bigcup_{p \in [\mathsf{K}]} \bigcup_{\{q \colon \mathbf{M}[p,q]=1\}} \hat{a}_{p,q} \neq a_{p,q} \Bigg] \tag{6}$$

tends to 0 as the blocklength $\mathsf{T} \to \infty$. The set of all achievable rate tuples is called the *capacity region* $\mathcal{C}(\mathsf{P})$.

Our main interest is in the Sum-DoF:

$$\text{Sum-DoF} \triangleq \varlimsup_{\mathsf{P} \to \infty} \sup_{\mathbf{R} \in \mathcal{C}(\mathsf{P})} \sum_{(p,q) \colon \mathbf{M}[p,q]=1} \frac{\mathsf{R}_{p,q}}{\log \mathsf{P}}. \tag{7}$$

Remark 1. *By definition, the* Sum-DoF *remains unchanged if one relabels the set of transmitters or the set of receivers with an arbitrary permutation of* $\{1, \ldots, \mathsf{K}\}$. *As a consequence, permuting the rows or the columns of the matrices* $\mathbf{M}$ *and* $\mathbf{N}$ *in the same way does not change the* Sum-DoF *of the network.*

III. Main Results

We first present the number-filling puzzle

A. The Puzzle

Consider a network with connectivity matrix $\mathbf{N}$ and message flow matrix $\mathbf{M}$. Such a network is called an $(\mathbf{M}, \mathbf{N})$-network. The puzzle is described by a K-by-K matrix $\mathbf{G}$ with non-negative integers, i.e., elements of $\mathbb{N}_0$.

Definition 1. *For a given connectivity and message flow matrices* $\mathbf{M}, \mathbf{N} \in \{0,1\}^{\mathsf{K} \times \mathsf{K}}$, *a matrix* $\mathbf{G} \in \mathbb{N}_0^{\mathsf{K} \times \mathsf{K}}$ *is called a* ***valid precoding index matrix*** *if the following two requirements are satisfied:*

R1) If $\mathbf{M}[p,q] = 0$ *or* $\mathbf{N}[p,q] = 0$ *then* $\mathbf{G}[p,q] = 0$*;*

R2) If $\mathbf{G}[p,q] = \mathbf{G}[p',q]$ *for* $p \neq p'$, *then* $\mathbf{G}[p,q] = 0$

Denote $\mathcal{G}_{\mathbf{M},\mathbf{N}}$ the set of all valid precoding index matrices for given $\mathbf{M}$ and $\mathbf{N}$.

Also denote by $\mathbf{G}^{(p)}$ the submatrix of $\mathbf{G}$ obtained by removing the p-th row and the columns $q \in [\mathsf{K}]$ for which $\mathbf{N}[p,q] = 0$. More formally:

$$\mathbf{G}^{(p)} \triangleq \mathbf{G}[p' \neq p, \{q \colon \mathbf{N}[p,q] = 1\}]. \tag{8}$$

(Notice the dependence of $\mathbf{G}^{(p)}$ on the connectivity matrix $\mathbf{N}$, which is not represented in the notation.) Let $g^{(p)}$ be equal to the number of different nonzero integers in the matrix $\mathbf{G}^{(p)}$, which is

$$g^{(p)} \triangleq \left| \left\{ \mathbf{G}^{(p)}[p',q] \neq 0 \colon p', q \in [\mathsf{K}] \right\} \right| \tag{9}$$

The goal of the puzzle is to maximize the following score of the index matrix

$$\mathsf{S}(\mathbf{G}) \triangleq \frac{\|\mathbf{G}\|_0}{\max_{p \in [\mathsf{K}]} \left\{ \|\mathbf{G}[p,:]\|_0 + g^{(p)} \right\}} \tag{10}$$

where $\|\mathbf{G}[p,:]\|_0$ denotes the number of nonzero entries in the p-th row of $\mathbf{G}$.

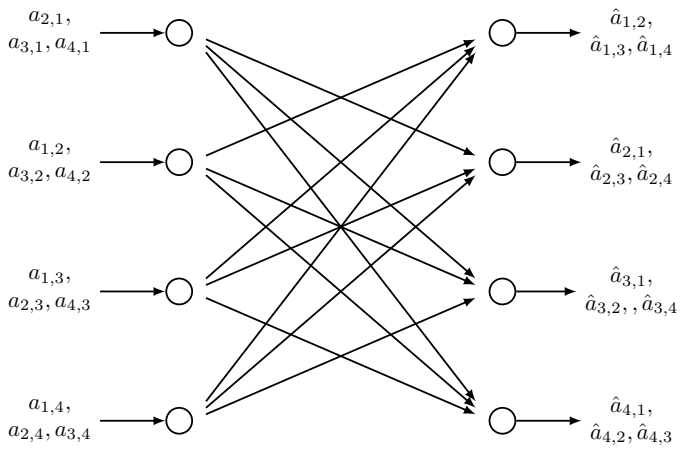


Fig. 1. An interference network where each Rx j receives a linear combination of signals transmitted by all Txs except Tx j, and each Tx k sends an independent message to every Rx except Rx k. The corresponding matrices $\mathbf{N}$ and $\mathbf{M}$ are given in (11).

Example 1. *We look at the example in Fig. 1, where the corresponding matrices* $\mathbf{M}$ *and* $\mathbf{N}$ *are given below.*

$$\mathbf{M} = \mathbf{N} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}. \tag{11}$$

This channel is similar to an X-channel, except there is one less link for each Rx. In this case, $\mathbf{G}^{(p)}$ *is obtained by removing the* p*-th row and* p*-th column in* $\mathbf{G}$*.*

Two possible fillings are

$$\mathbf{G} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 2 & 0 & 2 & 2 \\ 3 & 3 & 0 & 3 \\ 0 & 2 & 3 & 0 \end{bmatrix} \quad \text{or} \quad \mathbf{G} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 2 & 0 & 2 & 2 \\ 3 & 3 & 0 & 3 \\ 4 & 4 & 4 & 0 \end{bmatrix}. \tag{12}$$

We first examine the filling on the left. We find that $g^{(p)} = 2$ *for* $p \in \{1, 2, 3\}$ *as the integer* p *is not present in* $\mathbf{G}^{(p)}$*, whereas* $g^{(4)} = 3$*. Notice that* $\mathbf{G}$ *on the left has one fewer nonzero positive integer then the message flow matrix* $\mathbf{M}$ *(position marked in red), so* $\|\mathbf{G}[p, :]\|_0 = 3$ *for* $p \in \{1, 2, 3\}$ *and* $\|\mathbf{G}[4, :]\|_0 = 2$*, which yields the score* $\mathsf{S} = \frac{11}{5}$*. For the filling on the right, we have* $\|\mathbf{G}[p, :]\|_0 = 3$ *and* $g^{(p)} = 3$ *for every row. We obtain the score* $\mathsf{S} = 2$*, which is smaller than the score of the first filling. It is intriguing to discover that a higher score can be achieved by reducing the total number of positive integers in* $\mathbf{G}$*.*

Our first result in this paper connects the score $\mathsf{S}(\mathbf{G})$ to the Sum-DoF achieved by an IA scheme constructed from $\mathbf{G}$.

Proposition 1. *Given a* $(\mathbf{M}, \mathbf{N})$*-channel. Any* $\mathbf{G} \in \mathcal{G}_{\mathbf{M},\mathbf{N}}$ *corresponds to an IA scheme (with precoding matrices indicated by* $\mathbf{G}$*) that achieves a* Sum-DoF *equal to the score* $\mathsf{S}(\mathbf{G})$ *defined in* (10)

Proof. See Section IV. □

As described in Section IV, each entry in $\mathbf{G}$ corresponds to a transmitted message, and thus the numerator in (10) indicates the total number of transmitted messages. This explains Rule R1.), which requires that the entry in the puzzle be only nonzero if a message can be sent on the respective link. From Section IV it can further be seen that all messages corresponding to entries in $\mathbf{G}$ labeled with the same integer are transmitted using the same precoding matrix. Rule R2 ensures that any transmitter uses each precoding matrix for the transmission of only one of its messages, which is required to ensure that the different desired signal spaces at the receivers can be separated. The denominator in the score (10) captures the total number of signal dimensions required at the most constrained receiver, where $g^{(p)}$ describes the dimension of the aligned interference space and $\|\mathbf{G}[p, :]\|_0$ the dimension of the desired signal space.

Definition 2. *Given an* $(\mathbf{M}, \mathbf{N})$*-channel, the IA scheme corresponding to precoding index matrix*

$$\mathbf{G}_{\text{stand}}[p, q] = \begin{cases} p, & \text{if } \mathbf{N}[p, q] \cdot \mathbf{M}[p, q] = 1 \\ 0 & \text{otherwise} \end{cases} \tag{13}$$

is called the standard IA scheme*. This definition is motivated by the canonical interference alignment scheme of Cadambe and Jafar [7].*

B. Bounds on Sum-DoF: General Networks

Based on Proposition 1, we immediately obtain the following theorem.

Theorem 1. *For an $(\mathbf{M},\mathbf{N})$-channel, the* Sum-DoF *is lower bounded by the maximum score* S *of our puzzle:*

$$\text{Sum-DoF} \geq \text{Sum-DoF}_{\text{IA}} \triangleq \max_{\mathbf{G}\in\mathcal{G}_{\mathbf{M},\mathbf{N}}} \mathsf{S}(\mathbf{G}). \tag{14}$$

Though for general networks the lower bound in Theorem 1 is not known to be tight, it is the best available lower bound to date. Its computation involves solving a discrete optimization problem, which is generally hard. The task is simplified by noting that the problem can be phrased as a $0-1$ integer program, for which standard numerical tools exist.

Proposition 2. *The largest* Sum-DoF *achieved with IA,* $\text{Sum-DoF}_{\text{IA}}$ *is the solution of the following* $0-1$ *integer program:*

$$\text{Sum-DoF}_{\text{IA}} = \max_{\tilde{\mathbf{G}},\boldsymbol{w},z} \frac{\sum_{p,q,i=1}^{\mathsf{K}} \tilde{\mathbf{G}}[p,q,i]}{z} \tag{15a}$$

$$\text{s.t.} \sum_{i=1}^{\mathsf{K}} \tilde{\mathbf{G}}[p,q,i] \leq 1, \qquad \forall (p,q)\in[\mathsf{K}]^2, \tag{15b}$$

$$\sum_{i=1}^{\mathsf{K}} \tilde{\mathbf{G}}[p,q,i] = 0, \qquad \text{if } \mathbf{N}[p,q]\cdot\mathbf{M}[p,q]=0 \tag{15c}$$

$$\sum_{p=1}^{\mathsf{K}} \tilde{\mathbf{G}}[p,q,i] \leq 1, \qquad \forall (q,i)\in[\mathsf{K}]^2, \tag{15d}$$

$$\sum_{p',q=1}^{\mathsf{K}} \tilde{\mathbf{G}}^{(p)}[p',q,i] \leq \mathsf{K}^2 w_i^{(p)}, \ \ \forall (p,i)\in[\mathsf{K}]^2, \tag{15e}$$

$$\sum_{q,i=1}^{\mathsf{K}} \tilde{\mathbf{G}}[p,q,i] + \sum_{i=1}^{\mathsf{K}} w_i^{(p)} \leq z, \forall p\in[\mathsf{K}], \tag{15f}$$

$$\tilde{\mathbf{G}}[p,q,i],\ w_i^{(p)} \in \{0,1\}, \qquad \forall (p,q,i)\in[\mathsf{K}]^3. \tag{15g}$$

Our next result is an upper bound on the Sum-DoF. It is stated as a linear program (LP) and thus simple to evaluate numerically.

Theorem 2 (Converse)**.** *For an $(\mathbf{M},\mathbf{N})$-channel, the Sum-DoF is upper bounded as*

$$\text{Sum-DoF}(\mathbf{M},\mathbf{N}) \leq \mathsf{C}_{\text{dual}}(\mathbf{M},\mathbf{N}), \tag{16}$$

where $\mathsf{C}_{\text{dual}}(\mathbf{M},\mathbf{N})$ is the optimal value of the following linear program:

$$\mathsf{C}_{\text{dual}}(\mathbf{M},\mathbf{N}) = \min_{\{\lambda_{pq}\}} \sum_{p=1}^{\mathsf{K}}\sum_{q=1}^{\mathsf{K}} \lambda_{pq} \tag{17a}$$

$$\text{s.t.} \sum_{p=1}^{\mathsf{K}} \lambda_{pk} + \sum_{q=1}^{\mathsf{K}} \lambda_{jq} - \lambda_{jk} \geq 1, \forall (j,k): M[j,k]=1, \tag{17b}$$

$$\lambda_{pq} \geq 0, \quad \forall p,q\in[\mathsf{K}], \tag{17c}$$

$$\lambda_{pq} = 0, \quad \forall (p,q): \mathbf{N}[p,q]=0. \tag{17d}$$

Proof. The converse proof is presented in Section V and extends the argument in [3], [6], [14]. The proof first derives a collection of upper bounds on DoFs of subsets of messages by applying MAC-type arguments to sets of suitably chosen sub-channels. By taking linear combinations of these inequalities, an upper bound is then obtained on the Sum-DoF of the network. The optimization problem in above theorem stems from the set of possible choices of the linear combinations. □

Remark 2. *The upper bound C_{dual} is expressed as the solution of a linear program (LP), which can efficiently be solved numerically. Moreover, any feasible choice of the optimization parameters $\{\lambda_{pq}\}$ yields an upper bound on C_{dual} and thus on* Sum-DoF.

C. Bounds on Sum-DoF: Regular Networks

In this section, we focus on a class of regular networks and message flows.

Specifically, assume that Rx p only receives a signal from Tx q if $(p-q) \bmod \mathsf{K} < n$, where n is a given positive integer, and each Tx q only sends an independent message to Rx p if $(p-q) \bmod \mathsf{K} < m$, for $m \leq n$ another given positive integer. In this case,

$$\mathbf{N}[p,q] = \begin{cases} 1 & \text{if } (p-q) \bmod \mathsf{K} < n \\ 0 & \text{otherwise} \end{cases}. \tag{18a}$$

$$\mathbf{M}[p,q] = \begin{cases} 1 & \text{if } (p-q) \bmod \mathsf{K} < m \\ 0 & \text{otherwise} \end{cases}. \tag{18b}$$

The class of our regular networks includes the standard interference channel ($n = \mathsf{K}$ and $m = 1$), the standard X-channel ($n = m = \mathsf{K}$), and after relabeling of transmitters or receivers (see Remark 1), the *off-diagonal X-channel* in which the matrices $\mathbf{M}$ and $\mathbf{N}$ have all one-entries, except for zero diagonal entries as in Equation (11) in Example 1.

The following theorem proposes a lower bound on the Sum-DoF for the general class of regular networks based on a specific precoding index matrix $\mathbf{G}$, and shows that it coincides or is close to the upper bound in Theorem 2.

Theorem 3. *Consider a regular $(\mathbf{M}, \mathbf{N})$-channel as in* (18a) *and* (18b)*.*

For $m = n = 1$, we have:

$$\text{Sum-DoF} = \mathsf{K}. \tag{19}$$

For $n > m \geq 1$,

$$\text{Sum-DoF} = \frac{\mathsf{K}}{2}. \tag{20}$$

For $m = n > 1$,

$$\text{Sum-DoF} \geq \frac{\mathsf{K}m - (\mathsf{K} \bmod m) + \epsilon}{2m-1}, \tag{21}$$

where

$$\epsilon = \begin{cases} 1, & m = (\mathsf{K}+1)/2 \text{ and } \mathsf{K} \text{ is odd}, \\ 0, & \text{otherwise}, \end{cases} \tag{22}$$

and

$$\text{Sum-DoF} \leq \frac{\mathsf{K} \cdot m}{2m-1}. \tag{23}$$

Proof: Converse results follow from Theorem 2; details are presented in Section VI-A.

For $m = n = 1$, the transmission is interference-free, and Sum-DoF is thus equal to K.

For $n > m \geq 1$, the lower bound follows by Proposition 1 with the following choices of precoding index matrices:

$$\mathbf{G}[p,q] = \begin{cases} 1 + \big((p-q) \bmod \mathsf{K}\big), & \mathbf{M}[p,q] = 1, \\ 0, & \text{otherwise}. \end{cases} \tag{24}$$

For $n = m > 1$, the lower bound follows by Proposition 1 with the following choices of precoding index matrices. When $\epsilon = 0$ we choose:

$$\mathbf{G}[p,q] = \begin{cases} p \bmod_1 m, & 1 \leq p \leq K_1,\ \mathbf{M}[p,q] = 1, \\ m + q - p, & K_1 < p \leq \mathsf{K}, \\ & \quad p - m + 1 \leq q \leq p - 1, \\ 0, & \text{otherwise}, \end{cases} \tag{25}$$

where $\mathsf{K}_1 \triangleq \lfloor \mathsf{K}/m \rfloor \cdot m$.

When $\epsilon = 1$ and $\mathsf{K} = 3$, we choose

$$\mathbf{G} = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 1 & 0 \\ 0 & 2 & 3 \end{bmatrix}. \tag{26}$$

Finally, when $\epsilon = 1$ and $\mathsf{K} \geq 4$, we choose

$$\mathbf{G}[p,q] = \begin{cases} p, & 1 \leq p \leq m-1,\ \mathbf{M}[p,q] = 1, \\ m, & p = m,\ \mathbf{M}[m,q] = 1,\ q \neq m, \\ 1, & p = q = m, \\ m+q-p, & m+1 \leq p \leq \mathsf{K}-1, \\ & p-m+1 \leq q \leq p-2, \\ m-1, & m+1 \leq p \leq \mathsf{K}-1,\ q = p, \\ m, & p = \mathsf{K},\ \mathbf{M}[\mathsf{K},q] = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{27}$$

Section VI-B and Section VI-C derive the score for the proposed $\mathbf{G}$ matrices. ■

Figure 2 compares the upper and lower bounds in Theorem 3 for the regular channel and $\mathsf{K} = 20$. The bounds are also compared to the Sum-DoF achieved by the standard IA scheme [6] (corresponding to $\mathbf{G}_{\text{stand}}$ in (13)), which equals Sum-DoF of $\mathsf{K} \cdot m/(m + \min\{\mathsf{K}-1, 2m-2\})$, see Appendix B. We observe that upper and lower bounds in Theorem 3 are very close and considerably improve over the standard IA scheme.

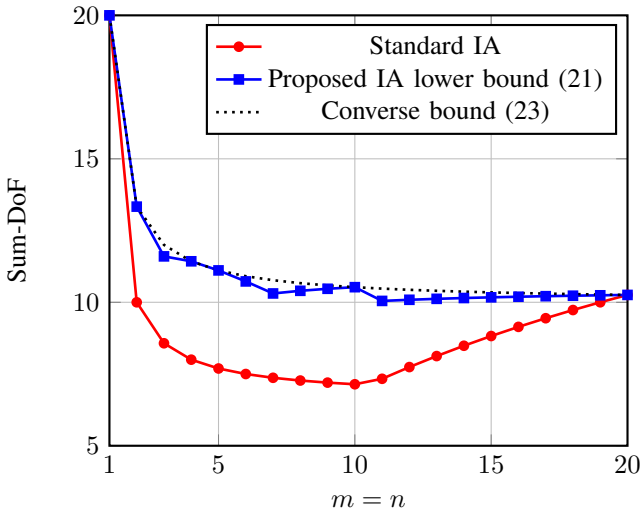


Fig. 2. Comparison of the Sum-DoF bounds from Theorem 3 with the standard IA scheme for a regular network with $\mathsf{K} = 20$ and $m = n$.

Theorem 3 recovers as special cases:

- the exact Sum-DoF $= \frac{\mathsf{K}}{2}$ of the standard interference channel ($n = \mathsf{K}$ and $m = 1$) [3];
- the exact Sum-DoF $= \frac{\mathsf{K}^2}{2\mathsf{K}-1}$ of the standard X-channel ($n = m = \mathsf{K}$) [6];
- the lower bound on the Sum-DoF $\geq \frac{\mathsf{K}\cdot(\mathsf{K}-1)-1}{2\mathsf{K}-3}$ of the off-diagonal X-channel ($m = n = \mathsf{K}-1$) [14].

Corollary 1 (Tightness of the bounds in Theorem 3)**.** *Theorem 3 characterizes the exact Sum-DoF whenever $m = n = 1$ or $n > m$. For $m = n > 1$, its upper and lower bounds coincide if and only if*

$$m \mid \mathsf{K} \qquad \text{or} \qquad (\mathsf{K}, m) = (3, 2). \tag{28}$$

In all other cases with $m = n > 1$, the difference between the upper and lower bounds in Theorem 3 is

$$\frac{(\mathsf{K} \bmod m) - \epsilon}{2m-1}. \tag{29}$$

For $m = n > 1$ the score achieved by our constructions in (25) and (27) do not match the upper bound in Theorem 3. Numerical results (based on the integer program in Proposition 2) show however that their scores coincide with Sum-DoF$_{\text{IA}}$, i.e., that the proposed matrices $\mathbf{G}$ in (25) and (27) achieve optimal puzzle score. For all values of $\mathsf{K} \leq 20$ and $m = n > 1$.

For off-diagonal X-channels, we can prove optimality of the proposed matrices $\mathbf{G}$ in (25) and (27) even analytically.

Proposition 3. *For off-diagonal X-channels the lower bound in* (21) *coincides with the optimum puzzle score* Sum-DoF$_{\text{IA}}$*:*

$$\text{Sum-DoF}_{\text{IA}} = \max_{\mathbf{G} \in \mathcal{G}_{\mathbf{M},\mathbf{N}}} \mathsf{S}(\mathbf{G}) = \frac{\mathsf{K}(\mathsf{K}-1)-1}{2\mathsf{K}-3}. \tag{30}$$

Proof: For $\mathsf{K} = 2$ and $\mathsf{K} = 3$, optimality follows directly from Corollary 1. The case $\mathsf{K} \geq 4$ is proved in Appendix C. ■

Remark 3 (Relation to Map-Reduce Distributed Computing)**.** *For off-diagonal X-channels, we had already proposed the IA scheme corresponding to the matrix* $\mathbf{G}$ *in* (25) *in our work on wireless Map-Reduce distributed computing systems [19]. In fact, in a Map-Reduce framework where each file can be stored at exactly one node*[1]*, a symmetric and regular file assignment [19], [21] naturally leads to the same communication model as in off-diagonal* X*-channels. Proposition 3 shows that under this regular file assignment, the IA scheme employed for the shuffle-phase in [19] is first-order high-SNR optimal*[2] *within the IA framework considered in this paper.*

IV. Proof of Proposition 1

Fix an $(\mathbf{M}, \mathbf{N})$-channel and let $\mathbf{G}$ be a valid index precoding matrix for this channel. In the following, we describe an AI coding scheme achieving a Sum-DoF equal to the puzzle score $\mathsf{S}(\mathbf{G})$.

A. Coding Scheme

We fix a parameter $\eta \in \mathbb{Z}^+$ and define

$$p_{\max} \triangleq \arg\max_{p \in [\mathsf{K}]} \left\{ \|\mathbf{G}[p,:]\|_0 + g^{(p)} \right\}. \tag{31}$$

Choose

$$\mathsf{T} = \eta^{\Gamma} \|\mathbf{G}[p_{\max},:]\|_0 + (\eta+1)^{\Gamma} g^{(p_{\max})}. \tag{32}$$

where we specify the value of Γ later in Eq. (37). We only send messages $\{a_{p,q}\}$ with $\mathbf{G}[p,q] > 0$. Each message is encoded using a circularly symmetric Gaussian codebook of average power $\mathsf{P}/\|\mathbf{G}[:,q]\|_0$ and codeword length η^{Γ}. Each codeword is sent over a block of T consecutive channel uses. More precisely, let $\{\mathbf{b}_{p,q}\}$ denote the η^{Γ}-length codeword symbol for message $a_{p,q}$. Each Tx q forms its inputs as:

$$\mathbf{X}_q = \sum_{\{p:\ \mathbf{G}[p,q]>0\}} \mathbf{U}_{\mathbf{G}[p,q]} \mathbf{b}_{p,q}, \tag{33}$$

where matrices $\{\mathbf{U}_g\}_{g \in [g_{\max}]}$ are described shortly, and $g_{\max}$ is the largest element in $\mathbf{G}$. Notice that messages $a_{p,q}$ and $a_{p',q'}$ are actually multiplied by the same precoding matrix if $\mathbf{G}[p,q] = \mathbf{G}[p',q'] > 0$.

We recall that Rx p only receives signals from connected Txs, i.e. from Tx q with $\mathbf{N}[p,q] = 1$, which allows to write the observed signal at each Rx $p \in [\mathsf{K}]$ as:

$$\begin{aligned} \mathbf{Y}_p = & \underbrace{\sum_{\{q:\ \mathbf{G}[p,q]>0\}} \mathbf{H}_{p,q} \mathbf{U}_{\mathbf{G}[p,q]} \mathbf{b}_{p,q}}_{\text{desired signal}} \\ & + \underbrace{\sum_{\{q:\ \mathbf{N}[p,q]=1\}} \sum_{\substack{\{p':\ p' \neq p \\ \mathbf{G}[p',q]>0\}}} \mathbf{H}_{p,q} \mathbf{U}_{\mathbf{G}[p',q]} \mathbf{b}_{p',q}}_{\text{Interference}} + \mathbf{Z}_p, \end{aligned} \tag{34}$$

where $\mathbf{H}_{p,q} \triangleq \mathrm{diag}([H_{p,q}(1), H_{p,q}(2) \cdots H_{p,q}(\mathsf{T})])$, $\mathbf{Y}_p \triangleq (Y_p(1), \ldots, Y_p(\mathsf{T}))^T$, and $\mathbf{Z}_p$ are the corresponding Gaussian noise vectors observed at Rx p.

B. IA Matrices $\{\mathbf{U}_g\}$

Inspired by the IA scheme in [7], we choose each $\mathsf{T} \times \eta^{\Gamma}$ precoding matrix $\mathbf{U}_g$ so that its column-span includes all power products (with powers from 1 to η) of the channel matrices $\mathbf{H}_{p,q}$ that premultiply $\mathbf{U}_g$ in (34) when the coded symbol is treated as interference for the receiver. Defining for $g \in [g_{max}]$:

$$\begin{aligned} \mathcal{H}_g \triangleq \{ & \mathbf{H}_{p,q} \colon \mathbf{N}[p,q] = 1 \text{ and } \exists p' \in [\mathsf{K}] \backslash \{p\} \\ & \text{such that } \mathbf{G}[p',q] = g \}, \end{aligned} \tag{35}$$

we have

$$\mathbf{U}_g = \left[\prod_{\mathbf{H} \in \mathcal{H}_g} \mathbf{H}^{\alpha_{g,\mathbf{H}}} \cdot \mathbf{\Xi}_g \colon \ \forall \boldsymbol{\alpha}_g \in [\eta]^{\Gamma} \right], \tag{36}$$

[1]Map-Reduce systems where files can be assigned to multiple nodes were also addressed in [19], but using different techniques.

[2]More precisely, in [19], the first-order high-SNR performance is measured in terms of *Normalized Delivery-Time (NDT)*. The connection between Sum-DoF as studied in this paper and NDT is made precise in [19, Lemma 1].

where $\{\mathbf{\Xi}_g\}_{g\in\mathbf{G}}$ are i.i.d. random vectors independent of all channel matrices, noises, and messages, and $\boldsymbol{\alpha}_g$ denotes a vector of the form $\boldsymbol{\alpha}_g = (\alpha_{g,\mathbf{H}}\colon \quad \mathbf{H}\in\mathcal{H}_g)$ with $\alpha_{g,\mathbf{H}}\in[\eta]$. We assume that the sizes of the sets $\mathcal{H}_g$ for $g\in[g_{\max}]$ are all equal (otherwise we simply add additional i.i.d. diagonal random matrices, which does not change the rest of the proof) and define

$$\Gamma \triangleq |\mathcal{H}_g|. \tag{37}$$

C. Analysis of Signal- and Interference-Subspaces

Since the column-span of $\mathbf{U}_g$ contains all power products of powers 1 to η of the channel matrices $\mathbf{H}_{p,q}$ that premultiply $\mathbf{U}_g$ in (34), the product of any of these matrices with $\mathbf{U}_g$ is included in the column-space of the $\mathsf{T}\times(\eta+1)^\Gamma$-matrix

$$\mathbf{W}_g = \left[\prod_{\mathbf{H}\in\mathcal{H}_g}\mathbf{H}^{\tilde{\alpha}_{g,\mathbf{H}}}\cdot\mathbf{\Xi}_g\colon\ \ \forall\tilde{\boldsymbol{\alpha}}_g\in[\eta+1]^\Gamma\right], \qquad g\in[g_{\max}]. \tag{38}$$

Formally, for each $g\in[g_{\max}]$ and $\mathbf{H}\in\mathcal{H}_g$, we have $\text{span}(\mathbf{H}\cdot\mathbf{U}_g)\subseteq\text{span}(\mathbf{W}_g)$. As a consequence, the signal and interference space at a Rx $p\in[\mathsf{K}]$ is represented by the matrix:

$$\mathbf{\Lambda}_p \triangleq [\mathbf{D}_p, \mathbf{I}_p]\,. \tag{39}$$

with the signal subspaces given by the $\mathsf{T}\times\|\mathbf{G}[p,:]\|_0\eta^\Gamma$-matrices

$$\mathbf{D}_p \triangleq \left[\mathbf{H}_{p,q}\mathbf{U}_{\mathbf{G}[p,q]}\colon \mathbf{G}[p,q]>0\right]. \tag{40}$$

and the interference subspaces given by

$$\mathbf{I}_p \triangleq \left[\mathbf{W}_{\mathbf{G}[p',q]}\colon \mathbf{N}[p,q]=1, p'\neq p, \mathbf{G}[p',q]>0\right]. \tag{41}$$

We can make the following observation:

Property 1. *For matrix $\mathbf{H}_{p,q}\mathbf{U}_{\mathbf{G}[p,q]}$ in the signal subspace $\mathbf{D}_p$, we have $\mathbf{H}_{p,q}\notin\mathcal{H}_{\mathbf{G}[p,q]}$ according to the second condition on valid precoding index matrices $\mathbf{G}$.*

Proof: The property can be proven by contradiction. We assume $\mathbf{H}_{p,q}\in\mathcal{H}_{\mathbf{G}[p,q]}$. By the definition of $\mathcal{H}$ in (35), we deduce that $\exists p'\in[\mathsf{K}]\backslash\{p\}$ such that $\mathbf{G}[p',q]=\mathbf{G}[p,q]$. There is thus at least one nonzero integer which repeats within a column of $\mathbf{G}$. But this violates the second requirement of a valid precoding index matrix in Definition 1, and yields the desired contradiction. ∎

The above property is used to prove the next property:

Property 2. *Each matrix $\{\mathbf{\Lambda}_p\}$ is of full column rank with probability 1.*

Property 2 implies that each Rx p can separate its desired signals from each other and from the interference space $\mathbf{I}_p$ at this Rx. Since the signal space $\mathbf{D}_p$ consists of $\|\mathbf{G}[p,:]\|_0\eta^\Gamma$ columns. In the limits $\eta\to\infty$ (and thus $\mathsf{T}\to\infty$) and $\mathsf{P}\to\infty$, Rx p achieves a DoF of

$$\lim_{\eta\to\infty}\frac{\|\mathbf{G}[p,:]\|_0\eta^\Gamma}{\mathsf{T}} = \frac{\|\mathbf{G}[p,:]\|_0}{\|\mathbf{G}[p_{\max},:]\|_0+g^{(p_{\max})}}. \tag{42}$$

. The Sum-DoF is therefore given by the puzzle score $\mathsf{S}(\mathbf{G})$. This establishes Proposition 1.

Proof of Property 2: The proof is obtained by proving that the matrix $\mathbf{\Lambda}_p$ has the same form as the matrix $\mathbf{A}$ in Lemma 1 and satisfies the two conditions mentioned in this lemma. Trivially, then any square submatrix of $\mathbf{\Lambda}_p$ also has this form and by Lemma 1 is full rank with probability 1. Since $\mathbf{\Lambda}_p$, is a tall matrix, we conclude that it is full column rank with probability 1.

To see that $\mathbf{\Lambda}_p$ is of the form in (43), notice that all matrices involved in (39), i.e. $\{\mathbf{H}_{p,q}\}$, are diagonal, and their multiplications with a vector $\mathbf{\Xi}_g$ from the right lead to column-vectors with t-components equal to the slot-t channel coefficient of $\{H_{p,q}(t)\}$ times the t-th element of the vector $\mathbf{\Xi}_g$. By definition, all t-th entries are independent of each other and of all random variables in the other rows. Therefore, the matrix $\mathbf{\Lambda}_p$ satisfies Condition i) in Lemma 1. To see that it also satisfies Condition ii), notice that for any two distinct columns $\mathbf{v}^{(1)}$ and $\mathbf{v}^{(2)}$ selected from $\mathbf{\Lambda}_p$, the exponents in the corresponding columns differ because:

1) If $\mathbf{v}^{(1)}$ and $\mathbf{v}^{(2)}$ are selected from the same signal subspace $\mathbf{H}_{p,q}\mathbf{U}_g$ or the same interference subspace $\mathbf{W}_g$, the two vectors have different exponents due to the construction method of $\mathbf{U}$ and $\mathbf{W}$.
2) If $\mathbf{v}^{(1)}$ is selected from the signal subspace $\mathbf{H}_{p,q}\mathbf{U}_g$, and $\mathbf{v}^{(2)}$ is selected from the signal subspace $\mathbf{H}_{p,q'}\mathbf{U}_{g'}$ or interference subspace $\mathbf{W}_{g'}$ with $g\neq g'$, the two vectors have different exponents as they have distinct factors $\mathbf{\Xi}_g$ and $\mathbf{\Xi}_{g'}$.

3) If $\mathbf{v}^{(1)}$ is selected from the signal subspace $\mathbf{H}_{p,q}\mathbf{U}_g$, and $\mathbf{v}^{(2)}$ is selected from the signal subspace $\mathbf{H}_{p,q'}\mathbf{U}_g$ or interference subspace $\mathbf{W}_g$, we can deduce has any factor $\mathbf{H}_{p,q} \in \mathcal{H}_g$ while by Property 1 that $\mathbf{v}^{(1)}$ $\mathbf{v}^{(2)}$ does not. The two vectors thus have different exponents.

This concludes the proof. ∎

Lemma 1 (Lemma 1 in [7])**.** *Consider an* F*-by-*F *square matrix* $\boldsymbol{A}$ *with* i*-th row and* j*-th column entry*

$$a_{ij} = \prod_{\ell=1}^{\mathsf{L}} \left(X_i^{[\ell]}\right)^{\alpha_{ij}^{[\ell]}}, \qquad i,j \in \mathsf{F}, \tag{43}$$

for random variables $\{X_i^{[\ell]}\}_{\ell\in[\mathsf{L}]}$ *and exponents*

$$\boldsymbol{\alpha}_{ij} \triangleq \left(\alpha_{ij}^{[1]}, \alpha_{ij}^{[2]}, \ldots, \alpha_{ij}^{[\mathsf{L}]}\right) \in \mathbb{Z}^{+\mathsf{L}}. \tag{44}$$

If

1) for any two pairs $(i,\ell) \neq (i',\ell')$ *the conditional cumulative probability distribution* $P_{X_i^{[\ell]}|X_{i'}^{[\ell']}}$ *is continuous; and*

2) any pair of vectors $\boldsymbol{\alpha}_{i,j} \neq \boldsymbol{\alpha}_{i,j'}$ *for* $i,j,j' \in [\mathsf{F}]$ *with* $j \neq j'$*;*

then the matrix $\boldsymbol{A}$ *is full rank with probability 1.*

V. Converse proof for Theorem 2

We first prove a useful lemma, and then conclude the converse proof by taking weighted sums of the upper bounds in the lemma.

A. A Useful Lemma

Lemma 2. *Let* $(\mathsf{R}_{p,q}(\mathsf{P})\colon \mathbf{M}[p,q]=1)$ *be a rate-tuple in* $\mathcal{C}(\mathsf{P})$*, for each* $\mathsf{P}>0$*. Then, for any* $j,k \in [\mathsf{K}]$ *with* $\mathbf{N}[j,k]=1$*:*

$$\overline{\lim_{\mathsf{P}\to\infty}} \left[\sum_{p\colon \mathbf{M}[p,k]=1} \frac{\mathsf{R}_{p,k}}{\log \mathsf{P}} + \sum_{q\colon \mathbf{M}[j,q]=1} \frac{\mathsf{R}_{j,q}}{\log \mathsf{P}} - \frac{\mathsf{R}_{j,k}}{\log \mathsf{P}} \right] \leq 1. \tag{45}$$

Proof: Fix $\mathsf{P}>0$ and any rate tuple $(\mathsf{R}_{p,q}(\mathsf{P})\colon \mathbf{M}[p,q]=1)$ in $\mathcal{C}(\mathsf{P})$. Then consider a sequence of encoding and decoding functions $\{f_q^{(\mathsf{T})}\}$ and $\{g_{p,k}^{(\mathsf{T})}\}$ such that $p^{(\mathsf{T})}(\text{error})$ tends to 0 as $\mathsf{T}\to\infty$.

Fix a blocklength T and indices $j,k \in [\mathsf{K}]$ with $\mathbf{N}[j,k]=1$, and define

$$\mathcal{F} \triangleq \{p\colon \mathbf{N}[p,k]=1 \text{ and } p\neq j\}. \tag{46}$$

Partition the set of messages into the following three sets

$$\mathbf{a}_r \triangleq \{a_{j,q}\colon \mathbf{M}[j,q]=1\}, \tag{47}$$

$$\mathbf{a}_t \triangleq \{a_{p,k}\colon \mathbf{M}[p,k]=1 \text{ and } p\neq j\}, \tag{48}$$

$$\mathbf{a}_c \triangleq \{a_{p,q}\colon \mathbf{M}[p,q]=1\} \setminus (\mathbf{a}_r \cup \mathbf{a}_t). \tag{49}$$

Finally, denote by $\mathcal{H}$ the set of *all* channel coefficients in the system, and for any subset $\mathcal{S}\subseteq[\mathsf{K}]$ define $\mathbf{Y}_{\mathcal{S}} \triangleq \left(Y_p^{(T)}\colon p\in\mathcal{S}\right)$ and $\mathbf{Z}_{\mathcal{S}} \triangleq \left(Z_p^{(T)}\colon p\in\mathcal{S}\right)$.

Notice now that by the independence of the messages, the channel coefficients, and the noise sequences:

$$\mathsf{T}\left(\sum_{p\colon \mathbf{M}[p,k]=1} \mathsf{R}_{p,k} + \sum_{q\colon \mathbf{M}[j,q]=1} \mathsf{R}_{j,q} - \mathsf{R}_{j,k}\right) \tag{50}$$

$$= H(\mathbf{a}_r, \mathbf{a}_t) \tag{51}$$

$$= H(\mathbf{a}_t, \mathbf{a}_r | \mathbf{a}_c, \mathcal{H}) \tag{52}$$

$$= I(\mathbf{a}_t, \mathbf{a}_r; \mathbf{Y}_j | \mathbf{a}_c, \mathcal{H}) + H(\mathbf{a}_t, \mathbf{a}_r | \mathbf{a}_c, \mathbf{Y}_j, \mathcal{H}) \tag{53}$$

$$= h(\mathbf{Y}_j|\mathbf{a}_c,\mathcal{H}) - h(\mathbf{Z}_j) + H(\mathbf{a}_r|\mathbf{a}_c,\mathbf{Y}_j,\mathcal{H}) + H(\mathbf{a}_t|\mathbf{a}_r,\mathbf{a}_c,\mathbf{Y}_j,\mathcal{H}) \tag{54}$$

$$\leq \mathsf{T}\cdot\log\left(1+\mathsf{P}\mathsf{H}_{\max}^2(\mathsf{K})\right) + H(\mathbf{a}_r|\mathbf{a}_c,\mathbf{Y}_j,\mathcal{H}) + H(\mathbf{a}_t|\mathbf{a}_r,\mathbf{a}_c,\mathbf{Y}_j,\mathcal{H}). \tag{55}$$

If communication is reliable, it is possible to reconstruct $\mathbf{a}_r$ from $\mathbf{Y}_j$ with probability of error tending to 0 as $\mathsf{T} \to \infty$. Therefore, by Fano's inequality

$$H(\mathbf{a}_r|\mathbf{a}_c, \mathbf{Y}_j, \mathcal{H}) \leq \mathsf{T} \cdot \epsilon_{\mathsf{T}}, \tag{56}$$

for some sequence $\{\epsilon_{\mathsf{T}}\}$ tending to 0 as $\mathsf{T} \to \infty$.

To bound the last summand in (55), we further notice that for reliable communication, Fano's inequality also implies

$$H(\mathbf{a}_t|\mathbf{a}_r, \mathbf{a}_c, \mathbf{Y}_j, \mathbf{Y}_{\mathcal{F}}, \mathcal{H}) \leq \mathsf{T}\tilde{\epsilon}_{\mathsf{T}}, \tag{57}$$

for some sequence $\{\tilde{\epsilon}_{\mathsf{T}}\}$ tending to 0 as $\mathsf{T} \to \infty$. Thus,

$$\begin{aligned}
&H(\mathbf{a}_t|\mathbf{a}_r, \mathbf{a}_c, \mathbf{Y}_j, \mathcal{H}) \\
&\leq H(\mathbf{a}_t|\mathbf{a}_r, \mathbf{a}_c, \mathbf{Y}_j, \mathcal{H})] - H(\mathbf{a}_t|\mathbf{a}_r, \mathbf{a}_c, \mathbf{Y}_j, \mathbf{Y}_{\mathcal{F}}, \mathcal{H}) + \mathsf{T}\tilde{\epsilon}_{\mathsf{T}} \\
&= I(\mathbf{a}_t; \mathbf{Y}_{\mathcal{F}}|\mathbf{a}_r, \mathbf{a}_c, \mathbf{Y}_j, \mathcal{H}) + \mathsf{T}\tilde{\epsilon}_{\mathsf{T}} && (58)\\
&\overset{(a)}{=} h(\tilde{\mathbf{Y}}_{\mathcal{F}}|\mathbf{a}_r, \mathbf{a}_c, \tilde{\mathbf{Y}}_j, \mathcal{H}) - h(\mathbf{Z}_{\mathcal{F}}) + \mathsf{T}\tilde{\epsilon}_{\mathsf{T}} && (59)\\
&\overset{(b)}{=} \mathbb{P}(E=1) \cdot h(\tilde{\mathbf{Y}}_{\mathcal{F}}|\mathbf{a}_r, \mathbf{a}_c, \tilde{\mathbf{Y}}_j, \mathcal{H}, E=1) \\
&\quad + \mathbb{P}(E=0) \cdot h(\tilde{\mathbf{Y}}_{\mathcal{F}}|\mathbf{a}_r, \mathbf{a}_c, \tilde{\mathbf{Y}}_j, \mathcal{H}, E=0) \\
&\quad - h(\mathbf{Z}_{\mathcal{F}}) + \mathsf{T}\tilde{\epsilon}_{\mathsf{T}} && (60)\\
&\overset{(c)}{=} h(\tilde{\mathbf{Y}}_{\mathcal{F}}|\mathbf{a}_r, \mathbf{a}_c, \tilde{\mathbf{Y}}_j, \mathcal{H}, E=1) - h(\mathbf{Z}_{\mathcal{F}}) + \mathsf{T}\tilde{\epsilon}_{\mathsf{T}} && (61)\\
&\overset{(d)}{\leq} h(\mathbf{Z}'_{\mathcal{F}}|\mathbf{a}_r, \mathbf{a}_c, \tilde{\mathbf{Y}}_j, \mathcal{H}, E=1) - h(\mathbf{Z}_{\mathcal{F}}) + \mathsf{T}\tilde{\epsilon}_{\mathsf{T}} && (62)\\
&\overset{(e)}{\leq} h(\mathbf{Z}'_{\mathcal{F}}) - h(\mathbf{Z}_{\mathcal{F}}) + \mathsf{T} \cdot \tilde{\epsilon}_{\mathsf{T}}, && (63)
\end{aligned}$$

where

- in (a) we defined for any $\mathcal{S} \subseteq [\mathsf{K}]$ the tuple $\tilde{\mathbf{Y}}_{\mathcal{S}} \triangleq \big(\tilde{Y}_p^{(T)} \colon p \in \mathcal{S}\big)$ with $\tilde{Y}_p^{(T)} \triangleq \big(\tilde{Y}_p(1), \ldots, \tilde{Y}_p(\mathsf{T})\big)$ and
$$\tilde{Y}_p(t) \triangleq H_{p,k}(t)X_k(t) + Z_p(t), \quad p \in \mathcal{S}; \tag{64}$$
- in (b) we defined the random variable E equal to 1 if the matrix $\mathbf{H}_{j,k}$ is invertible. In this case, input $X_k(t)$, for $t \in [\mathsf{T}]$, can be obtained by the entries in $\tilde{\mathbf{Y}}_j - \mathbf{Z}_j$.
- (c) holds because $h(\tilde{\mathbf{Y}}_{\mathcal{F}}|\mathbf{a}_r, \mathbf{a}_c, \tilde{\mathbf{Y}}_j, \mathcal{H}, E=0)$ is bounded and because $\mathbb{P}(E=1) = 1$. This latter fact holds because $E = 1$ whenever a specific square matrix of the random channel coefficients is invertible (see (b) above), which happens with probability 1 since the channel coefficients are independently drawn from a continuous distribution;
- in (d) we defined the tuple $\mathbf{Z}_{\mathcal{F}} \triangleq (Z'_p(t) \colon p \in \mathcal{F},\ t \in [\mathsf{T}])$ and
$$Z'_p(t) \triangleq Z_p(t) - H_{p,k}(t)(H_{j,k}(t))^{-1}(Y_j(t) - Z_j(t)), \tag{65}$$
- in (e) we used the independence of the noise from the channel coefficients and the fact that conditioning can only decrease differential entropy.

Finally, combining Eqs. (55), (56), and (63), we obtain

$$\begin{aligned}
&\sum_{p\colon \mathbf{M}[p,k]=1} \mathsf{R}_{p,k} + \sum_{q\colon \mathbf{M}[j,q]=1} \mathsf{R}_{j,q} - \mathsf{R}_{j,k} && (66)\\
&\leq \log\big(1 + \mathsf{P}\mathsf{H}_{\max}^2(\mathsf{K})\big) + \epsilon_{\mathsf{T}} \\
&\quad + \frac{1}{\mathsf{T}} h(\mathbf{Z}'_{\mathcal{F}}) - \frac{1}{\mathsf{T}} h(\mathbf{Z}_{\mathcal{F}}) + \tilde{\epsilon}_{\mathsf{T}}. && (67)
\end{aligned}$$

Letting $\mathsf{T} \to \infty$ and $\mathsf{P} \to \infty$ establishes the desired inequality in the lemma, because $h(\mathbf{Z}'_{\mathcal{F}})/\mathsf{T}$ and $h(\mathbf{Z}_{\mathcal{F}})/\mathsf{T}$ are both finite constants that do not depend on T nor P, and both sequences ϵ_{T} and $\tilde{\epsilon}_{\mathsf{T}}$ tend to 0 as $\mathsf{T} \to \infty$. ■

B. Concluding the Dual Linear Program

To simplify the notation, we define

$$d_{ij} = \overline{\lim_{\mathsf{P}\to\infty}} \frac{\mathsf{R}_{i,j}}{\log \mathsf{P}}. \tag{68}$$

Our auxiliary Lemma 2 implies the following upper bound:

$$\text{Sum-DoF}(\mathbf{M}, \mathbf{N}) \leq \mathsf{C}_{\text{primal}}(\mathbf{M}, \mathbf{N}) \tag{69}$$

where

$$\mathsf{C}_{\text{primal}}(\mathbf{M},\mathbf{N}) = \max_{\{d_{ij}\}} \sum_{(i,j):\mathbf{M}[i,j]=1} d_{ij} \tag{70}$$

s.t.

$$\sum_{i:\mathbf{M}[i,q]=1} d_{iq} + \sum_{j:\mathbf{M}[p,j]=1} d_{pj} - d_{pq} \le 1,$$

$$\forall (p,q): \quad \mathbf{N}[p,q] = 1 \tag{70a}$$

$$d_{ij} \ge 0, \quad \forall (i,j): \quad \mathbf{M}[i,j] = 1. \tag{70b}$$

The covering problem in (17) is the dual of the LP in (70) [22]. Whenever the LP in (69) is feasible and bounded, we can thus conclude that the upper bound in the theorem holds.

Remark 4. *The primal LP in* (70) *and the dual LP in* (17) *have comparable formulations and computational complexity. The dual LP however has the advantage of being a minimization problem, implying that each set of feasible variables provides an upper bound on the* Sum-DoF.

VI. Proof of Theorem 3

A. Converse Proofs

For case $m = n$, the proof follows immediately by summing the upper bound in Lemma 2 for the $\mathsf{K} \cdot m$ distinct pairs $(p,q) \in [\mathsf{K}] \times [\mathsf{K}]$ with $\mathbf{M}[p,q] = 1$, and then dividing this sum by $2m-1$, because each rate has been counted $2m-1$ times.

The upper bound can also be obtained directly from Theorem 2 by choosing

$$\lambda_{pq} = \frac{1}{2m-1}, \qquad \forall (p,q) \colon \mathbf{M}[p,q] = 1 \tag{71}$$

as a suboptimal feasible solution of the linear program.

For the case $m < n$, we sum up the upper bound in Lemma 2 for only pairs connected by an interfering link, which means the pairs (p,q) with $\mathbf{N}[p,q] = 1$ and $\mathbf{M}[p,q] = 0$. We can obtain the ratio $\mathsf{K}/2$ as there are $\mathsf{K} \cdot (n-m)$ pairs and each rate has been counted $2(n-m)$ times.

The same upper bound can be obtained from Theorem 2 with the choice

$$\lambda_{pq} = \frac{1}{2(n-m)}, \qquad \forall (p,q) \colon \mathbf{M}[p,q] = 0 \text{ and } \mathbf{N}[p,q] = 1. \tag{72}$$

B. Achievability Proof for $n > m \ge 1$

For $n > m \ge 1$, consider the following precoding index matrix:

$$\mathbf{G}[p,q] = \begin{cases} 1 + \big((p-q) \bmod \mathsf{K}\big), & \mathbf{M}[p,q] = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{73}$$

We first verify that $\mathbf{G}$ is a valid precoding index matrix. Since $m < n$, we have $\mathbf{M}[p,q] = 1$ only if $\mathbf{N}[p,q] = 1$, and thus Requirement R1) is satisfied. Moreover, for any fixed column q, the m nonzero entries correspond to the m distinct values

$$(p-q) \bmod \mathsf{K} \in 0, \ldots, m-1, \tag{74}$$

and are therefore assigned the distinct indices $1, \ldots, m$. Hence, no positive integer appears more than once in the same column, and Requirement R2) is also satisfied.

We next compute the score of $\mathbf{G}$. Each row of $\mathbf{G}$ contains exactly m nonzero entries, and therefore

$$|\mathbf{G}|_0 = \mathsf{K}m, \qquad |\mathbf{G}[p,:]|_0 = m, \quad \forall p \in [\mathsf{K}]. \tag{75}$$

It remains to compute $g^{(p)}$. We show that every integer in $[m]$ appears in $\mathbf{G}^{(p)}$. By the circular symmetry of $\mathbf{M}$, $\mathbf{N}$, and $\mathbf{G}$, it suffices to consider $\mathbf{G}^{(1)}$. Since $n > m$, the column $q = 1 - m \bmod_1 \mathsf{K}$ is retained in $\mathbf{G}^{(1)}$. Before removing the first row, the m nonzero entries in this column are assigned the indices $1, \ldots, m$. Moreover, none of these entries lies in the first row. Hence, all indices in $[m]$ appear in $\mathbf{G}^{(1)}$, and therefore

$$g^{(p)} = g^{(1)} = m. \tag{76}$$

Consequently,

$$|\mathbf{G}[p,:]|_0 + g^{(p)} = 2m, \qquad \forall p \in [\mathsf{K}], \tag{77}$$

and the resulting puzzle score is

$$\mathsf{S}(\mathbf{G}) = \frac{\mathsf{K}m}{2m} = \frac{\mathsf{K}}{2}. \tag{78}$$

By Proposition 1, this establishes the lower bound

$$\text{Sum-DoF} \geq \frac{\mathsf{K}}{2}. \tag{79}$$

Together with the converse bound proved in Section VI-A, we conclude that

$$\text{Sum-DoF} = \frac{\mathsf{K}}{2}. \tag{80}$$

C. Achievability Proof for $m = n > 1$

As we prove in this section, the scores of the matrices $\mathbf{G}$ proposed in (25) and (27) coincide with the lower bound in (21). We present the proofs for the cases $\epsilon = 0$ and $\epsilon = 1$ separately. For convenience of the reader, we also recall the proposed precoding index matrices in (25) and (27) in a different form and present examples for the score calculations.

1) The case $\epsilon = 0$: The general philosophy behind the construction in (25) is to have a first set of rows where all nonzero entries have same index in $[m]$ and a second set of rows where diagonal entries are set to 0 and all other entries that can be nonzero are assigned the numbers in $[m-1]$ in increasing order, up to a circular shift.

So, for rows $p = 1, \ldots, \lfloor \mathsf{K}/m \rfloor \cdot m$, we choose

$$\mathbf{G}[p,q] = \begin{cases} p \bmod_1 m & \text{if } \mathbf{M}[p,q] = 1 \\ 0 & \text{otherwise} \end{cases}, \tag{81a}$$

and for the remaining rows $p = \lfloor \mathsf{K}/m \rfloor \cdot m + 1, \ldots, \mathsf{K}$, we choose

$$\mathbf{G}[p,q] = \begin{cases} q - p \bmod m & \text{if } \mathbf{M}[p,q] = 1 \text{ and } p \neq q \\ 0 & \text{otherwise} \end{cases}. \tag{81b}$$

An example is presented in (85).

The proposed matrix $\mathbf{G}$ is a valid precoding index matrix because:

- $\mathbf{G}[p,q] > 0$ only when $\mathbf{M}[p,q] > 0$.
- Each positive number occurs only once in a column. This is trivial for integer m because it only occurs in row mg. To verify that an integer $i \in [m-1]$ occurs only once, notice that it appears in row i, at columns $i, i-1, i-2, \ldots, i-m+1$, where differences have to be understood $\bmod_1 \mathsf{K}$, and it also appears in each row $p \in \{\lfloor \mathsf{K}/m \rfloor m+1, \cdots, \mathsf{K}\}$ at column $p-m+i$. To sum up, integer $i \in [m-1]$ thus occurs in columns (up to $\bmod_1 \mathsf{K}$) $i, i-1, \ldots, i-m+1, p-m+1, \ldots, p-1$, which are all distinct numbers.

We proceed to analyze the score of the proposed $\mathbf{G}$. Since the diagonal entries of the last $\mathsf{K} \bmod m$ rows in $\mathbf{G}$ are set to 0 (although those entries in $\mathbf{M}$ are equal to 1), the total number of nonzero entries in G is:

$$\|\mathbf{G}\|_0 = \mathsf{K} \cdot m - (\mathsf{K} \bmod m). \tag{82}$$

To compute $g^{(p)}$ for all indices $p \in [\mathsf{K}]$, we distinguish between p smaller or larger than $\lfloor \mathsf{K}/m \rfloor \cdot m$. For $p \leq \lfloor \mathsf{K}/m \rfloor \cdot m$, the submatrix $\mathbf{G}^{(p)}$ does not include integer $p \bmod m$ but all other integer in $[m]$ and thus $g^{(p)} = m-1$. For $p > \lfloor \mathsf{K}/m \rfloor \cdot m$, the submatrix $\mathbf{G}^{(p)}$ includes all integers in $[m]$ and thus $g^{(p)} = m$. Since the first $\lfloor \mathsf{K}/m \rfloor \cdot m$ rows contain $\|\mathbf{G}[p,:]\|_0 = m$ nonzero symbols and the last $K \bmod m$ rows contain $\|\mathbf{G}[p,:]\|_0 = m-1$ nonzero symbols, we conclude that

$$\|\mathbf{G}[p,:]\|_0 + g^{(p)} = 2m - 1, \quad \forall p \in [\mathsf{K}], \tag{83}$$

is constant for all p. Combining (83) with (82) establishes the desired result in (21).

Example 2. *Let $\mathsf{K} = 6$ and $m = 4$, so that the connectivity and message flow matrices are:*

$$\mathbf{N} = \mathbf{M} = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}. \tag{84}$$

In this special case, the precoding index matrix $\mathbf{G}_{\epsilon=0}$ is:

$$\mathbf{G}_{\epsilon=0} = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 2 & 2 & 0 & 0 & 2 & 2 \\ 3 & 3 & 3 & 0 & 0 & 3 \\ 4 & 4 & 4 & 4 & 0 & 0 \\ 0 & 1 & 2 & 3 & 0 & 0 \\ 0 & 0 & 1 & 2 & 3 & 0 \end{bmatrix}, \tag{85}$$

where notice in the above we have $\lfloor \mathsf{K}/m \rfloor \cdot m = 4$ *and thus the first four rows have a unique entry, while the last two rows have* $m-1=3$ *entries and are circular shifts of each other.*

We count $\|\mathbf{G}\|_0 = 22$ *entries in total. Notice also* $\|\mathbf{G}[p,:]\|_0 = 4$*, for* $p \in \{1,2,3,4\}$*, and* $\|\mathbf{G}[p,:]\|_0 = 3$*, for* $p \in \{5,6\}$*. Then, consider the two submatrices* $\mathbf{G}^{(1)}$ *(*$\mathbf{G}^{(2)}$ *to* $\mathbf{G}^{(4)}$ *are similar) and* $\mathbf{G}^{(5)}$ *(*$\mathbf{G}^{(6)}$ *is similar) extracted from* (85)*:*

$$\mathbf{G}^{(1)} = \begin{bmatrix} 2 & 0 & 2 & 2 \\ 3 & 0 & 0 & 3 \\ 4 & 4 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 2 & 3 & 0 \end{bmatrix}, \mathbf{G}^{(5)} = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 2 & 0 & 0 & 2 \\ 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 \\ 0 & 1 & 2 & 3 \end{bmatrix} \tag{86}$$

As $\mathbf{G}^{(1)}$ *doesn't contain the integer* 1*, we obtain that* $g^{(1)} = 3$*, while* $g^{(5)} = 4$ *because* $\mathbf{G}^{(5)}$ *contains all indices in* $[4]$ *For all* $p \in [6]$*, we thus have* $\|\mathbf{G}[p,:]\|_0 + g^{(p)} = 7$*.*

Combining all these considerations results in the score $\mathsf{S}(\mathbf{G}) = 22/7$*.*

2) The case $\epsilon = 1$ *and* $\mathsf{K} > 4$ *:* In this case, $\mathsf{K} > 4$ odd and $m = \frac{\mathsf{K}+1}{2}$, i.e.,

$$\mathsf{K} = 2m - 1. \tag{87}$$

Reconsider the matrix $\mathbf{G}$ from the previous subsection that assigns integer p to all entries in row $p = 1, \ldots, m$ where the corresponding entry in $\mathbf{M}$ is nonzero and in rows $p = m+1, \ldots, 2m-1$ assigns indices $i \in [m-1]$ in a circular fashion to the entries where $\mathbf{M}$ is nonzero, however with zeros on the diagonal. Change now the matrix so that

1) The diagonal entry in row m is set to 1.
2) For rows $p = m+1, \ldots, 2m$ the entries $\mathbf{G}[p, p-1]$ and $\mathbf{G}[p,p]$ are exchanged compared to the previous construction, i.e., the zero is now on the first off-diagonal and not on the diagonal.
3) In row $\mathsf{K} = 2m-1$ we set $\mathbf{G}[\mathsf{K}, q] = m$ for all entries where $\mathbf{M}[\mathsf{K}, q] = 1$.

To summarize, see also (27), we choose for rows $p = 1, \ldots, m-1$:

$$\mathbf{G}[p,q] = \begin{cases} p & \text{if } \mathbf{M}[p,q] = 1 \\ 0 & \text{otherwise} \end{cases}, \tag{88a}$$

for row $p = m$:

$$\mathbf{G}[m,q] = \begin{cases} m & \text{if } \mathbf{M}[p,q] = 1 \text{ and } p \neq q \\ 1 & \text{if } p = q \\ 0 & \text{otherwise} \end{cases}, \tag{88b}$$

and for rows $p = m+1, \ldots, 2m-2$:

$$\mathbf{G}[p,q] = \begin{cases} q - p \bmod m & \text{if } \mathbf{M}[p,q] = 1 \text{ and } p \notin \{q, q+1\} \\ m-1 & \text{if } \mathbf{M}[p,q] = 1 \text{ and } p = q \\ 0 & \text{otherwise} \end{cases}. \tag{88c}$$

For the last row $p = 2m-1$, we choose:

$$\mathbf{G}[2m-1,q] = \begin{cases} m & \text{if } \mathbf{M}[p,q] = 1 \\ 0 & \text{otherwise} \end{cases}. \tag{88d}$$

See (93) for an example of the construction.

We now show that the proposed matrix $\mathbf{G}$ is a valid precoding index matrix. By construction, $\mathbf{G}[p,q] > 0$ only if $\mathbf{M}[p,q] = 1$. It remains to verify that no positive integer appears twice in the same column. The symbol 1, for example, appears in columns $\{1, \mathsf{K}, \mathsf{K}-1, \ldots, \mathsf{K}-m+2\} = \{1, m+1, \ldots, 2m-1\}$ in row 1 and in each column $q = 2, \ldots, m-1$ in row $p = m+q-1$ only. Additionally, symbol 1 also appears in column m in row m only. To conclude, symbol 1 appears exactly once in each column. The analogous reasoning holds for symbols $2, \ldots, m-1$ except that there is one column where they do not appear. Symbol m appears in columns $1, \ldots, m-1$ in row m and it appears columns $m, \ldots, 2m-1$ in rows $2m-1$. It does not occur in all other rows. So the symbol m appears exactly once in each column. This proves that $\mathbf{G}$ is a valid precoding index matrix, and it remains to compute the score of $\mathbf{G}$.

To this end, notice that the first m rows and the last row each contain m nonzero entries, contributing $m(m+1)$ entries in total. Rows $p \in \{m+1, \ldots, \mathsf{K}-1\}$ each contain $m-1$ nonzero entries, and there are $m-2$ such rows, contributing $(m-2)(m-1)$ entries. Therefore,

$$\|\mathbf{G}\|_0 = m(m+1) + (m-2)(m-1) = \mathsf{K}m - (m-2). \tag{89}$$

We now compute $g^{(p)}$. For rows $p \in \{m+1, \ldots, \mathsf{K}-1\}$, all symbols in $[m]$ appear in $\mathbf{G}^{(p)}$, hence $g^{(p)} = m$. Moreover, for these values of p we have $\|\mathbf{G}[p,:]\|_0 = m-1$. For the first m rows and the last row, $p \in \{1, \ldots, m, \mathsf{K}\}$, one symbol is missing from $\mathbf{G}^{(p)}$, so $g^{(p)} = m-1$. Moreover, $\|\mathbf{G}[p,:]\|_0 = m$. These findings can be combined to conclude

$$\|\mathbf{G}[p,:]\|_0 + g^{(p)} = 2m-1, \tag{90}$$

constant for all $p \in [\mathsf{K}]$. Finally, combining (89) and (90), we obtain that for the proposed precoding matrix $\mathbf{G}$ the desired score

$$\mathsf{S}(\mathbf{G}) = \frac{\mathsf{K}m - (m-2)}{2m-1}. \tag{91}$$

Example 3. *Consider the case* $\mathsf{K} = 7$ *and* $m = 4$*, which satisfies* K *odd and* $\mathsf{K} = 2m-1$*. The corresponding connectivity and message flow matrices are*

$$\mathbf{N} = \mathbf{M} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}. \tag{92}$$

The precoding index matrix in (88) *is:*

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 2 & 2 & 0 & 0 & 0 & 2 & 2 \\ 3 & 3 & 3 & 0 & 0 & 0 & 3 \\ 4 & 4 & 4 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 & 4 & 4 & 4 \end{bmatrix}. \tag{93}$$

There are 26 *nonzero entries in* $\mathbf{G}$*, and thus* $\|\mathbf{G}\|_0 = 26$*. Moreover,* $\|\mathbf{G}[p,:]\|_0 = 4$*, for* $p \in \{1,2,3,4,7\}$*, and* $\|\mathbf{G}[p,:]\|_0 = 3$*, for* $p \in \{5,6\}$*. Next consider the three submatrices* $\mathbf{G}^{(1)}$ *(matrices* $\mathbf{G}^{(2)}$*,* $\mathbf{G}^{(3)}$*, and* $\mathbf{G}^{(4)}$ *are similar),* $\mathbf{G}^{(5)}$ *(the matrix* $\mathbf{G}^{(6)}$ *is similar), and* $\mathbf{G}^{(7)}$ *extracted from* (93)*:*

$$\mathbf{G}^{(1)} = \begin{bmatrix} 2 & 0 & 2 & 2 \\ 3 & 0 & 0 & 3 \\ 4 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 4 & 4 & 4 \end{bmatrix}, \quad \mathbf{G}^{(5)} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 4 \end{bmatrix},$$

$$\mathbf{G}^{(7)} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 \\ 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 2 & 0 & 3 & 0 \end{bmatrix}. \tag{94}$$

In the submatrix $\mathbf{G}^{(1)}$*, the integer* 1 *does not appear, and therefore* $g^{(1)} = 3$*. A similar argument applies to* $\mathbf{G}^{(2)}$*,* $\mathbf{G}^{(3)}$*,* $\mathbf{G}^{(4)}$ *and* $\mathbf{G}^{(7)}$*, yielding* $g^{(2)} = g^{(3)} = g^{(4)} = g^{(7)} = 3$*. By contrast, all integers in* $[4]$ *appear in both* $\mathbf{G}^{(5)}$ *and* $\mathbf{G}^{(6)}$*, so* $g^{(5)} = g^{(6)} = 4$*. Therefore, for all* $p \in [7]$*, we have*

$$\|\mathbf{G}[p,:]\|_0 + g^{(p)} = 7, \tag{95}$$

and the resulting lower bound on Sum-DoF *is* $\frac{26}{7}$.

VII. Conclusion

In this paper, we studied the Sum-DoF of a K-user interference network with general connectivity and general message flows, described respectively by matrices $\mathbf{N}$ and $\mathbf{M}$. Motivated by the fact that the straightforward application of the standard IA can be suboptimal in sparsely connected networks, we proposed a number-filling puzzle depending on $\mathbf{M}$ and $\mathbf{N}$ so that any valid filling corresponds to an IA scheme for the $(\mathbf{M}, \mathbf{N})$-network and an appropriately defined score of the puzzle indicates the Sum-DoF achieved by the IA scheme. We further showed that finding the optimal puzzle score for given matrices $(\mathbf{M}, \mathbf{N})$ can be cast as a 0-1 integer program.

We also derived a general converse bound for arbitrary $(\mathbf{M}, \mathbf{N})$-channels. The converse is obtained by combining a family of MAC-type upper bounds through a linear program and is presented as a linear program minimization problem.

For a class of symmetric regular networks, we proposed explicit constructions of valid number-fillings to the puzzle. These constructions lead to closed-form achievable Sum-DoF expressions and avoid solving the integer program. In the case $m < n$, our result recovers the classical $\mathsf{K}/2$ Sum-DoF. In the case $m = n$, the proposed construction strictly improves upon the straightforward implementation of the standard IA, where all messages intended for the same receiver are assigned the same precoding matrix. Moreover, the achieved Sum-DoF is close to our upper bound. For the off-diagonal X-channel, we further showed that the proposed construction achieves optimal puzzle score.

Our results on symmetric regular networks also imply that the shuffle-phase IA-scheme for MapReduce systems with computation load $\mathsf{r} = 1$ proposed in [19] is first-order high-SNR optimal (i.e., achieves the smallest Normalized Delivery-Time) within all IA schemes of the framework considered in this paper.

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APPENDIX A
EXPRESSING THE OPTIMAL PUZZLE SCORE $\text{Sum-DoF}_{\text{IA}}$ AS AN INTEGER PROGRAM

We can naturally represent any valid index precoding matrix $\mathbf{G}$ by a binary matrix $\tilde{\mathbf{G}} \in \{0,1\}^{\mathsf{K}\times\mathsf{K}\times\mathsf{K}}$, by setting

$$\tilde{\mathbf{G}}[p,q,i] = 1 \Leftrightarrow \mathbf{G}[p,q] = i, \tag{96}$$

for any $(p,q,i) \in [\mathsf{K}]^3$.[3] Notice that since each entry of $\mathbf{G}$ contains only one integer, $\tilde{\mathbf{G}}$ satisfies the following exclusivity constraint:

$$\sum_{i=1}^{\mathsf{K}} \tilde{\mathbf{G}}[p,q,i] \leq 1, \quad \forall (p,q) \in [\mathsf{K}]^2. \tag{97}$$

Moreover, whenever $\mathbf{G}$ is a valid precoding index matrix, the inherited $\tilde{\mathbf{G}}$ must satisfy the following two conditions (inherited from Rules R1) and R2)):

$\tilde{\text{R}}1)$ $\sum_{i=1}^{\mathsf{K}} \tilde{\mathbf{G}}[p,q,i] = 0$ if $\mathbf{N}[p,q] = 0$ or $\mathbf{M}[p,q] = 0$.

$\tilde{\text{R}}2)$ $\sum_{p=1}^{\mathsf{K}} \tilde{\mathbf{G}}[p,q,i] \leq 1, \quad \forall (q,i) \in [\mathsf{K}]^2$.

Notice the next equalities

$$\|\mathbf{G}\|_0 = \sum_{p,q,i \in [\mathsf{K}]} \tilde{\mathbf{G}}[p,q,i] \tag{98}$$

and

$$\|\mathbf{G}[p,:]\|_0 = \sum_{q,i \in [\mathsf{K}]} \tilde{\mathbf{G}}[p,q,i], \tag{99}$$

and define

$$\tilde{\mathbf{G}}^{(p)}[:,:,i] = \mathbf{Id}_{p^c} \cdot \tilde{\mathbf{G}}[:,:,i] \cdot \text{diag}(\mathbf{N}[p,:]), \tag{100}$$

where $\mathbf{Id}_{p^c}$ denotes the identity matrix whose p-th diagonal entry is replaced by 0. This operation removes the contribution of row p and keeps only the columns selected by the neighborhood pattern $\mathbf{N}[p,:]$, and thus is the counterpart of $\mathbf{G}^{(p)}$.

It can now be concluded that Sum-DoF$_{\text{IA}}$ equals the solution of the $0-1$ integer program in (15) by combining (97)–(100) with the rules $\tilde{\text{R}}1)$ and $\tilde{\text{R}}2)$, and by noticing that the $0-1$ integer program in (15) has an optimal solution with slack variables $\{w_i^{(p)}\}$ and z satisfying the following conditions:

- $w_i^{(p)} = 0$ if, and only if, $\tilde{\mathbf{G}}^{(p)}[:,:,i]$ is the all-zero matrix, or equivalently if, and only if, $\mathbf{G}^{(p)}$ does not contain index i; otherwise $w_i^{(p)} = 1$. Notice that in this case:
$$\sum_{i=1}^{\mathsf{K}} w_i^{(p)} = g^{(p)}. \tag{101}$$
- The slack variable z equals $\max_{p \in [\mathsf{K}]} \sum_{q,i=1}^{\mathsf{K}} \tilde{\mathbf{G}}[p,q,i] + \sum_{i=1}^{\mathsf{K}} w_i^{(p)}$, and so by the previous item, (101) and (99):
$$z = \max_{p \in [\mathsf{K}]} \left(\|\mathbf{G}[p,:]\|_0 + g^{(p)} \right). \tag{102}$$

APPENDIX B
Sum-DoF ACHIEVED BY STANDARD IA

For a regular network with $m = n$, consider the straightforward implementation of the standard IA scheme, in which all messages intended for the same receiver are assigned the same precoding matrix. That means, consider the precoding index matrix satisfying

$$\mathbf{G}[p,q] = \begin{cases} p, & \text{if } \mathbf{M}[p,q] = 1, \\ 0, & \text{if } \mathbf{M}[p,q] = 0. \end{cases} \tag{103}$$

Since each row of $\mathbf{M}$ contains m ones, each row of $\mathbf{G}$ contains m nonzero entries. Therefore,

$$\|\mathbf{G}\|_0 = \mathsf{K} \cdot m, \quad \|\mathbf{G}[p,:]\|_0 = m, \qquad \forall p \in [\mathsf{K}]. \tag{104}$$

It remains to compute $g^{(p)}$. Recall that $\mathbf{G}^{(p)}$ is obtained by removing row p and retaining only the columns q with $\mathbf{N}[p,q] = 1$. In this case, the columns are $\{q\colon (p-q) \bmod \mathsf{K} < m\}$, which are $q \in p, p-1, \ldots, p-m+1$ where the indices are understood modulo K.

An integer $i \neq p$ appears in $\mathbf{G}^{(p)}$ if, and only if, there exists a retained column q such that

$$(i-q) \bmod \mathsf{K} < m. \tag{105}$$

Since the retained columns range from $p-m+1$ to p, the labels that can appear range from

$$p-m+1, \ldots, p-1, p+1, \ldots, p+m-1, \tag{106}$$

[3]Without loss in optimality we assume $i \in [\mathsf{K}]$. In fact, for any achievable score S, it is possible to find an index precoding matrix $\mathbf{G}$ with score $\mathsf{S}(\mathbf{G}) = \mathsf{S}$.

again modulo K. Consequently, the number of distinct nonzero labels in $\mathbf{G}^{(p)}$ is

$$g^{(p)} = \min\{\mathsf{K}-1, 2m-2\}, \quad \forall p \in [K]. \tag{107}$$

According to Proposition 1, the Sum-DoF achieved by this straightforward scheme is therefore

$$\mathsf{S}(\mathbf{G}) = \frac{\mathsf{K}\cdot m}{m + \min\{\mathsf{K}-1, 2m-2\}}. \tag{108}$$

APPENDIX C
PROOF OF PROPOSITION 3

Consider the off-diagonal X-channel with $\mathsf{K} \geq 4$. We prove that

$$\text{Sum-DoF}_{\text{IA}} = \frac{\mathsf{K}(\mathsf{K}-1)-1}{2\mathsf{K}-3}. \tag{109}$$

The proposed precoding index matrices $\mathbf{G}$ in (25) and (27) establish that $\text{Sum-DoF}_{\text{IA}} \geq \frac{\mathsf{K}(\mathsf{K}-1)-1}{2\mathsf{K}-3}$. It remains to prove:

$$\text{Sum-DoF}_{\text{IA}} \leq \frac{\mathsf{K}(\mathsf{K}-1)-1}{2\mathsf{K}-3}. \tag{110}$$

We start with some definitions and observations. Fix a valid precoding index matrix $\mathbf{G}$. Then, define for each positive integer $a \in \mathbb{Z}^+$, the set

$$\mathcal{E}_a \triangleq \{(i,j)\colon \mathbf{G}[i,j] = a\}, \tag{111}$$

and its number of occurrences in $\mathbf{G}$:

$$t_a \triangleq |\mathcal{E}_a|. \tag{112}$$

Since the same integer cannot appear twice in one column,

$$t_a \leq \mathsf{K}. \tag{113}$$

Define for each $k \in [\mathsf{K}]$ the cross

$$\mathcal{X}_k \triangleq \{(k,j)\colon j \in [\mathsf{K}]\} \cup \{(i,k)\colon i \in [\mathsf{K}]\}, \tag{114}$$

and observe that for any integer a,

$$a \notin \mathbf{G}^{(k)} \quad \Longleftrightarrow \quad \mathcal{E}_a \subseteq \mathcal{X}_k. \tag{115}$$

We are now ready to prove the desired inequality (110), i.e., that no precoding index matrix $\mathbf{G}$ has a score exceeding the right-hand side of (110). We separately consider the matrices $\mathbf{G}$ that have at least one row with $\mathsf{K}-1$ entries and the matrices where all rows have no more than $\mathsf{K}-2$ entries.

A. $\mathbf{G}$ *has a row with* $\mathsf{K}-1$ *entries*

Without loss of generality, suppose that the first row of $\mathbf{G}$ has $\mathsf{K}-1$ entries. Partition the matrix $\mathbf{G}$ as

$$\mathbf{G} = \begin{bmatrix} \times & \mathbf{G}_r \\ \mathbf{G}_c & \mathbf{G}^{(1)} \end{bmatrix}, \tag{116}$$

where

$$\begin{aligned} \mathbf{G}_r &= \mathbf{G}[1,\, 2:\mathsf{K}], \\ \mathbf{G}_c &= \mathbf{G}[2:\mathsf{K},\, 1], \\ \mathbf{G}^{(1)} &= \mathbf{G}[2:\mathsf{K},\, 2:\mathsf{K}]. \end{aligned} \tag{117}$$

By our assumption, $\mathbf{G}_r$ has all nonzero entries and thus:

$$\|\mathbf{G}[1,:]\| = \mathsf{K}-1. \tag{118}$$

Similarly, because integers cannot repeat in a column, and the submatrix $\mathbf{G}^{(1)}$ has at most $(\mathsf{K}-2)(\mathsf{K}-1)$ nonzero elements, we can bound

$$\|\mathbf{G}^{(1)}\|_0 \leq \min\{g^{(1)}, (\mathsf{K}-2)\}(\mathsf{K}-1). \tag{119}$$

where recall that $g^{(1)}$ denotes the number of distinct positive integers appearing in $\mathbf{G}^{(1)}$. Including also the elements in $\mathbf{G}_r$ and $\mathbf{G}_c$, we thus obtain:

$$\|\mathbf{G}\|_0 \leq \|\mathbf{G}^{(1)}\|_0 + (\mathsf{K}-1) + (\mathsf{K}-1) \tag{120}$$

$$\leq \min\{g^{(1)}, (\mathsf{K}-2)\}(\mathsf{K}-1) + 2(\mathsf{K}-1). \tag{121}$$

Let $r \geq 0$ be the integer value that satisfies

$$\|\mathbf{G}\|_0 \quad = \min\{g^{(1)} + 2, \mathsf{K}\}(\mathsf{K}-1) - r. \tag{122}$$

Moreover, let $d \geq 0$ be the integer value that satisfies

$$\mathsf{K} - 1 + g^{(1)} = \max_{p \in [\mathsf{K}]} \left\{ \|\mathbf{G}[p,:]\|_0 + g^{(p)} \right\} - d. \tag{123}$$

Above considerations allow us to upper bound the score of $\mathbf{G}$ as

$$\mathsf{S}(\mathbf{G}) \quad \leq \quad \frac{\min\{g^{(1)} + 2, \mathsf{K}\}(\mathsf{K}-1) - r}{(\mathsf{K}-1) + g^{(1)} + d} \tag{124}$$

$$\leq \quad \frac{\mathsf{K}(\mathsf{K}-1) - r}{2\mathsf{K} - 3 + d}, \tag{125}$$

where the last inequality holds because for $\mathsf{K} \geq 4$ the right-hand side of (124) is monotonically increasing for $g^{(1)} \leq \mathsf{K} - 2$ and decreasing for larger values of $g^{(1)}$. For $(r \geq 1, d \geq 0)$ and for $(r = 0, d \geq 1)$, the right-hand side of (125) trivially does not exceed the score on the right-hand side of (110). (In fact, for $(r = 0, d \geq 1)$, right-hand side of (125) evaluates to $(\mathsf{K}(\mathsf{K}-1))/(2\mathsf{K}-2) = \mathsf{K}/2$.)

We are left with checking the case $r = d = 0$, which implies that both (119) and (120) hold with equality, and moreover

$$\mathsf{K} - 1 + g^{(1)} \geq \|\mathbf{G}[p,:]\|_0 + g^{(p)}, \quad p \in [\mathsf{K}]. \tag{126}$$

We first consider the case $g^{(1)} \geq \mathsf{K} - 1$. Since $r = d = 0$, (124) gives

$$\mathsf{S}(\mathbf{G}) \leq \frac{\mathsf{K}(\mathsf{K}-1)}{(\mathsf{K}-1) + g^{(1)}} \leq \frac{\mathsf{K}}{2}, \tag{127}$$

which is smaller than the right-hand side of (110). It therefore remains to consider $g^{(1)} \leq \mathsf{K} - 2$, in which case equality in (119) implies that each of the $g^{(1)}$ positive integers in $\mathbf{G}^{(1)}$ appears exactly $\mathsf{K} - 1$ times.

Notice that equality in (120) implies that $\mathbf{G}_c$ has all nonzero entries, and since the entries in a column cannot repeat, $\mathbf{G}_c$ consists of $\mathsf{K} - 1$ different positive integers. As a consequence,

$$g^{(p)} \geq \mathsf{K} - 2, \quad \forall p \in \{2, \dots \mathsf{K}\}, \tag{128}$$

and by (126),

$$\|\mathbf{G}[p,:]\|_0 \leq g^{(1)} + 1, \quad \forall p \in \{2, \dots \mathsf{K}\}. \tag{129}$$

Moreover, to also have equality in (119), since $\mathbf{G}_c$ has all nonzero entries, we must have equality in (129). By (126) this implies also equality in (128).

We conclude this proof by showing that no valid precoding index matrix can satisfy the following three equalities:

$$g^{(p)} = \mathsf{K} - 2, \quad \forall p \in \{2, \dots \mathsf{K}\}, \tag{130a}$$

$$\|\mathbf{G}[p,:]\|_0 = g^{(1)} + 1, \quad \forall p \in \{2, \dots \mathsf{K}\} \tag{130b}$$

$$\|\mathbf{G}_r\|_0 = \|\mathbf{G}_c\|_0 = \mathsf{K} - 1 \tag{130c}$$

and one condition:

C1) every positive integer appearing in $\mathbf{G}^{(1)}$ has multiplicity $\mathsf{K} - 1$.

Without loss of generality, suppose that

$$\mathbf{G}_c = [2, 3, \dots, \mathsf{K}]^T. \tag{131}$$

By Condition (130a), this implies that for each $p \in \{2, \dots, \mathsf{K}\}$ the matrix $\mathbf{G}^{(p)}$ contains integers $[\mathsf{K}] \backslash \{1, p\}$ but no other positive integer. As a consequence, the entire matrix $\mathbf{G}$ only contains integers $2, \dots, \mathsf{K}$, and moreover by (115):

$$\mathcal{E}_p \subseteq \mathcal{X}_p. \tag{132}$$

Under these two conditions, the only possible filling of $\mathbf{G}_r$ is

$$\mathbf{G}_r = [2, 3, \dots, \mathsf{K}]. \tag{133}$$

So, we conclude that under conditions (130) and C1) that the matrix $\mathbf{G}$ must be of the form (here we show $\mathsf{K} = 6$):

$$\mathbf{G} = \begin{bmatrix} 0 & 2 & 3 & 4 & 5 & 6 \\ 2 & 0 & ? & ? & ? & ? \\ 3 & ? & 0 & ? & ? & ? \\ 4 & ? & ? & 0 & ? & ? \\ 5 & ? & ? & ? & 0 & ? \\ 6 & ? & ? & ? & ? & 0 \end{bmatrix}, \tag{134}$$

with appropriate entries "?" consisting only of

a subset of the integers $2, \ldots, \mathsf{K}$ and each such integer occurring exactly $\mathsf{K}-1$ times. Since each integer p, $\mathcal{E}_p \in \mathcal{X}_p$, and it appears in column p of the first row, it cannot appear in this column anymore. As part of $\mathbf{G}^{(1)}$, integer p can thus only appear in row p. Since for an off-diagonal channel there are however only $\mathsf{K}-2$ such row-p nonzero entries, integer p cannot occur $\mathsf{K}-1$ times. As a consequence, every entry of $\mathbf{G}^{(1)}$ must be zero. This yields $g^{(1)} = 0$, which turns the upper bound on the score in (124) into (using $r = d = 0$)

$$\mathsf{S}(\mathbf{G}) \leq 2. \tag{135}$$

This completes the proof for matrices with at least one row that contains $\mathsf{K}-1$ entries.

B. $\mathbf{G}$ *has no row with* $\mathsf{K}-1$ *entries*

Define also v_a as the number of indices $k \in [\mathsf{K}]$ for which $a \in \mathbf{G}^{(k)}$ and causes interference:

$$v_a \triangleq \sum_{k \in [\mathsf{K}]} \mathbf{1}\left\{a \in \mathbf{G}^{(k)}\right\}, \tag{136}$$

where $\mathbf{1}$ is the indicator function. By double counting,

$$\sum_{k=1}^{\mathsf{K}} g^{(k)} = \sum_{a \in \mathcal{A}} v_a. \tag{137}$$

By (115), we have

$$a \notin (\mathbf{G}^{(p)} \cup \mathbf{G}^{(q)}) \quad \Longleftrightarrow \quad \mathcal{E}_a \subseteq (\mathcal{X}_p \cap \mathcal{X}_q), \tag{138}$$

and as a consequence $t_a \leq 2$. Thus, if $v_a \leq \mathsf{K}-2$, then $t_a \leq 2$. If $v_a = \mathsf{K}-1$, then symbol a can only be in one of the crosses, and since it can only occur once in a column, and we assume that each row has at most $\mathsf{K}-2$ entries, we have $t_a \leq \mathsf{K}-1$. In the final case $v_a = \mathsf{K}$, we can conclude that $t_a \leq \mathsf{K}$ because each integer can occur at most once in each column. For $\mathsf{K} \geq 4$, above observations imply

$$v_a \geq t_a. \tag{139}$$

We use (137) to obtain

$$\sum_{k=1}^{\mathsf{K}} g^{(k)} = \sum_{a \in \mathbb{Z}^+} v_a \geq \sum_{a \in \mathbb{Z}^+} t_a = \|\mathbf{G}\|_0.$$

Consequently,

$$\mathsf{S}(\mathbf{G}) = \frac{\|\mathbf{G}\|_0}{\max_{p \in [\mathsf{K}]} (\|\mathbf{G}[p,:]\|_0 + g^{(p)})} \tag{140}$$

$$\leq \frac{\|\mathbf{G}\|_0}{\frac{1}{\mathsf{K}} \sum_{p \in [\mathsf{K}]} (\|\mathbf{G}[p,:]\|_0 + g^{(p)})} \tag{141}$$

$$= \frac{\|\mathbf{G}\|_0}{\frac{1}{\mathsf{K}} \left(\|\mathbf{G}\|_0 + \sum_{p \in [\mathsf{K}]} g^{(p)} \right)} \tag{142}$$

$$= \frac{\mathsf{K}}{2}. \tag{143}$$

Since for $\mathsf{K} \geq 2$:

$$\frac{\mathsf{K}}{2} \leq \frac{\mathsf{K}(\mathsf{K}-1)-1}{2\mathsf{K}-3}, \tag{144}$$

we obtain

$$\mathsf{S}(\mathbf{G}) \leq \frac{\mathsf{K}(\mathsf{K}-1)-1}{2\mathsf{K}-3}. \tag{145}$$